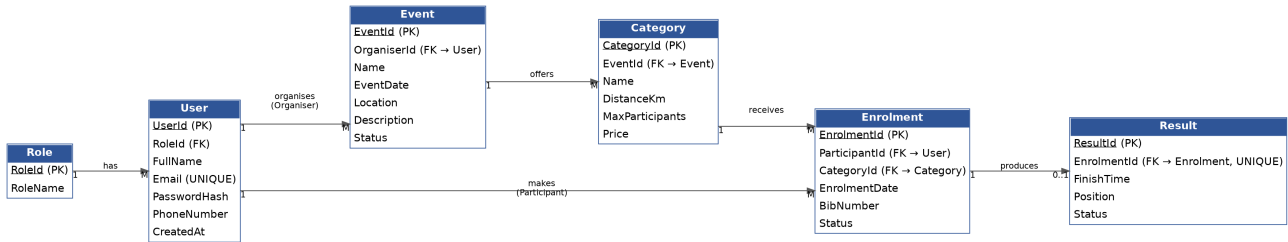


RaceDay System

Entity Relationship Diagram (ERD)



Entity Overview

Entity	Purpose	Primary Key
Role	Fixed lookup table of the two system roles.	RoleId
User	Every account: Organisers and Participants share one table, distinguished by RoleId.	UserId
Event	A race event created and owned by an Organiser (User).	EventId
Category	A race distance/division within an Event (e.g. 5km, 10km).	CategoryId
Enrolment	Links a Participant (User) to a Category they have signed up for.	EnrolmentId
Result	The finishing outcome for a single Enrolment.	ResultId

Relationships and Cardinality

Relationship	Cardinality	Meaning
Role → User	1 : M	Each role can apply to many users; each user has exactly one role.
User → Event	1 : M (Organiser)	An Organiser can create many events; each event has one owning Organiser.
Event → Category	1 : M	An event can offer many categories; each category belongs to one event.
User → Enrolment	1 : M (Participant)	A Participant can make many enrolments; each enrolment belongs to one participant.
Category → Enrolment	1 : M	A category can receive many enrolments; each enrolment is for one category.
Enrolment → Result	1 : 0..1	An enrolment produces at most one result, and only once the race has been run.

Design Notes

- A single **User** table holds both Organisers and Participants, distinguished via a normalised **Role** lookup table, rather than two separate people tables — this keeps role-based access control (who

can call which API endpoint) centred on one table.

- **BibNumber** is stored on **Enrolment**, not **Result**, because a bib is assigned at sign-up time, before the race happens.
- The **Enrolment** → **Result** relationship is 1:0..1 rather than a mandatory 1:1, since a result only exists once a participant has actually finished (or DNF'd/DQ'd) a race — this is enforced in SQL with a UNIQUE foreign key constraint on Result.EnrolmentId.
- This ERD contains six entities, meeting the assignment's minimum, and maps directly onto the six CREATE TABLE statements in the accompanying SQL script (raceday_schema.sql) and the endpoints in the API endpoint plan (api_endpoint_plan.md).